

Developing a Foundational Dataset for Urdu Text Summarization

Defining Research Questions, Design Thresholds, and a Conceptual Framework

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FAST NUCES Karachi, Pakistan

Date: December 4, 2025

The Research Problem: Bridging the Data Gap

Urdu, a language spoken by over 100 million people, faces significant challenges in the realm of Natural Language Processing (NLP). A critical impediment to progress in Urdu text summarization is the profound lack of a **foundational dataset and a robust conceptual framework**.

This absence severely restricts the development and evaluation of effective summarization models, hindering research and application development. Our work directly addresses this fundamental data gap.

Urdu's Unique Linguistic Challenges

Urdu presents a fascinating yet complex landscape for NLP due to its distinctive linguistic features, which pose considerable barriers to automated text summarization.



Rich Morphology

Complex word structures with numerous prefixes, suffixes, and infixes.



Free Word Order

Flexible sentence structure, complicating syntactic analysis.



Nastaliq Script

Flowing, cursive script requiring specialized rendering and processing.

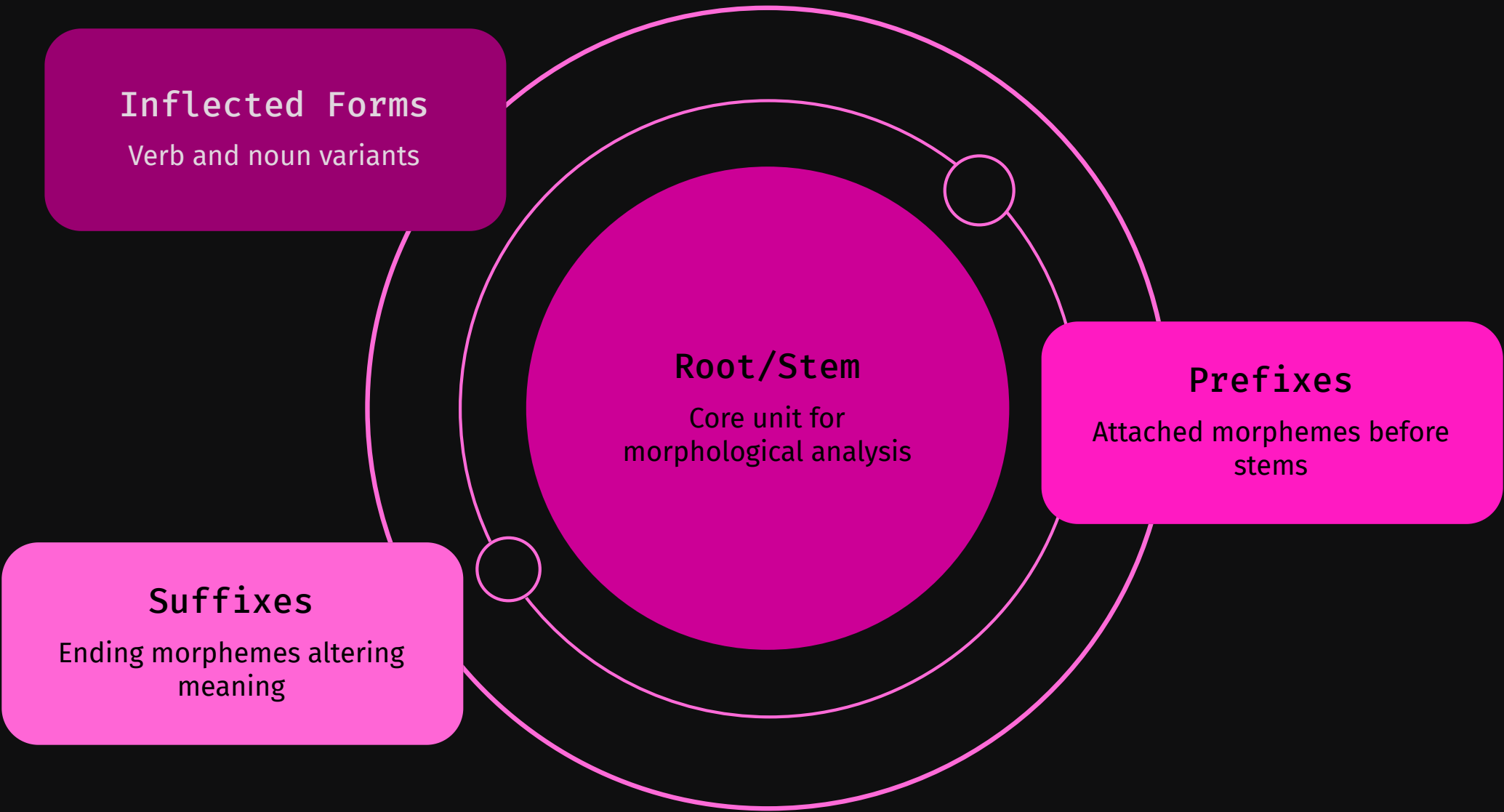
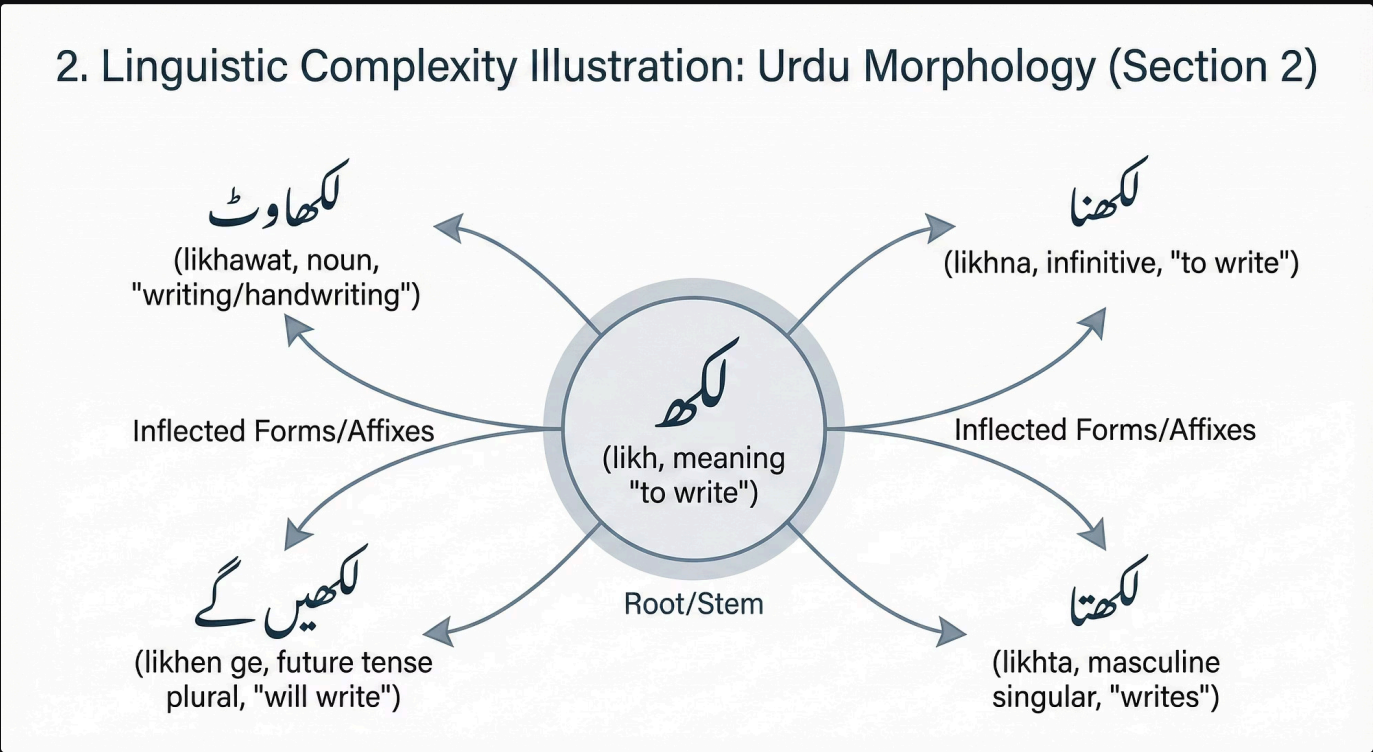


Perso-Arabic Vocabulary

Extensive loanwords from Persian and Arabic, impacting lexical resources.

Understanding the morphological complexity is key to developing effective NLP tools for Urdu.

2. Linguistic Complexity Illustration: Urdu Morphology (Section 2)



Summarization Context & Literature Review

Text summarization techniques are broadly categorized into two main approaches:

Extractive Summarization

- Identifies and extracts key sentences or phrases directly from the original text.
- Commonly uses methods like TF-IDF and TextRank.

Abstractive Summarization

- Generates new sentences and phrases to create a concise summary, often paraphrasing the original content.
- Leverages advanced Deep Learning models (e.g., Sequence-to-Sequence models).

For low-resource languages like Urdu, **Extractive Summarization serves as a pragmatic starting point**. Its reliance on identifying existing text segments makes it less dependent on vast amounts of annotated data required for abstractive models. ROUGE overlap is very high.

Core Contribution: A Blueprint for Data Infrastructure

Our primary contribution is not just a dataset, but a meticulously crafted **comprehensive conceptual framework** designed to guide the development of a foundational dataset for Urdu text summarization.

1

Systematic Approach

A structured methodology for dataset creation.

2

Design Thresholds

Clear, quantifiable metrics for quality assurance.

3

Scalability

Framework built to support expansion to larger datasets.

4

Reproducibility

Enables other researchers to build upon our work.

This framework acts as a definitive blueprint, providing the necessary guidelines for robust data infrastructure development, crucial for advancing Urdu NLP.

Framework Component 1: Source Acquisition

The foundation of any high-quality dataset lies in its source material. For Urdu text summarization, ensuring **domain diversity** is paramount to create a versatile and representative dataset.



News Articles

Reflects current events, factual reporting, and diverse topics.



Academic Texts

Introduces complex vocabulary and structured arguments. *Daryaft*, *Tehqeeq*, and *Journal of Research (Urdu)*.



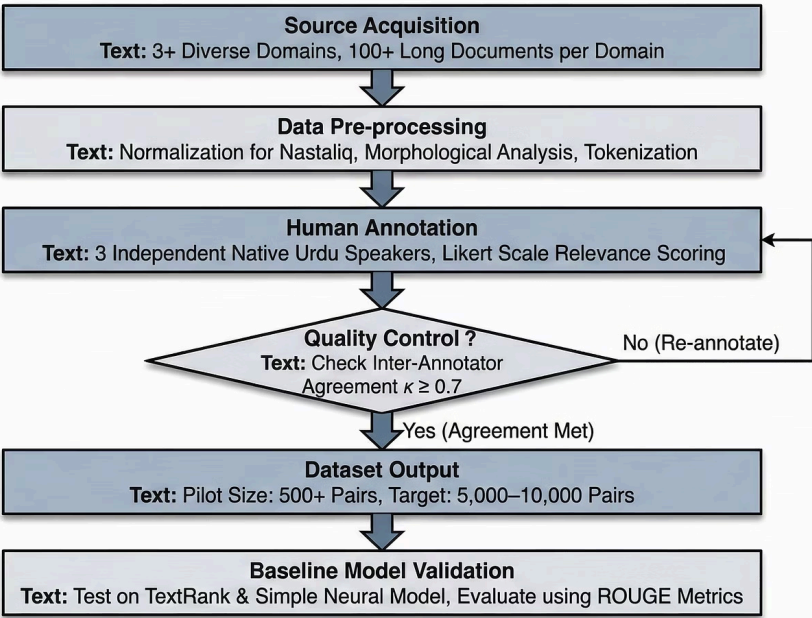
Literary Works

Captures nuanced language, figurative speech, and varied narrative styles. (esp Perso-Arabic vocabulary)

Design Threshold: The dataset must encompass **at least three distinct domains** to ensure comprehensive coverage and prevent domain-specific bias. This breadth is critical for developing generalized summarization models.

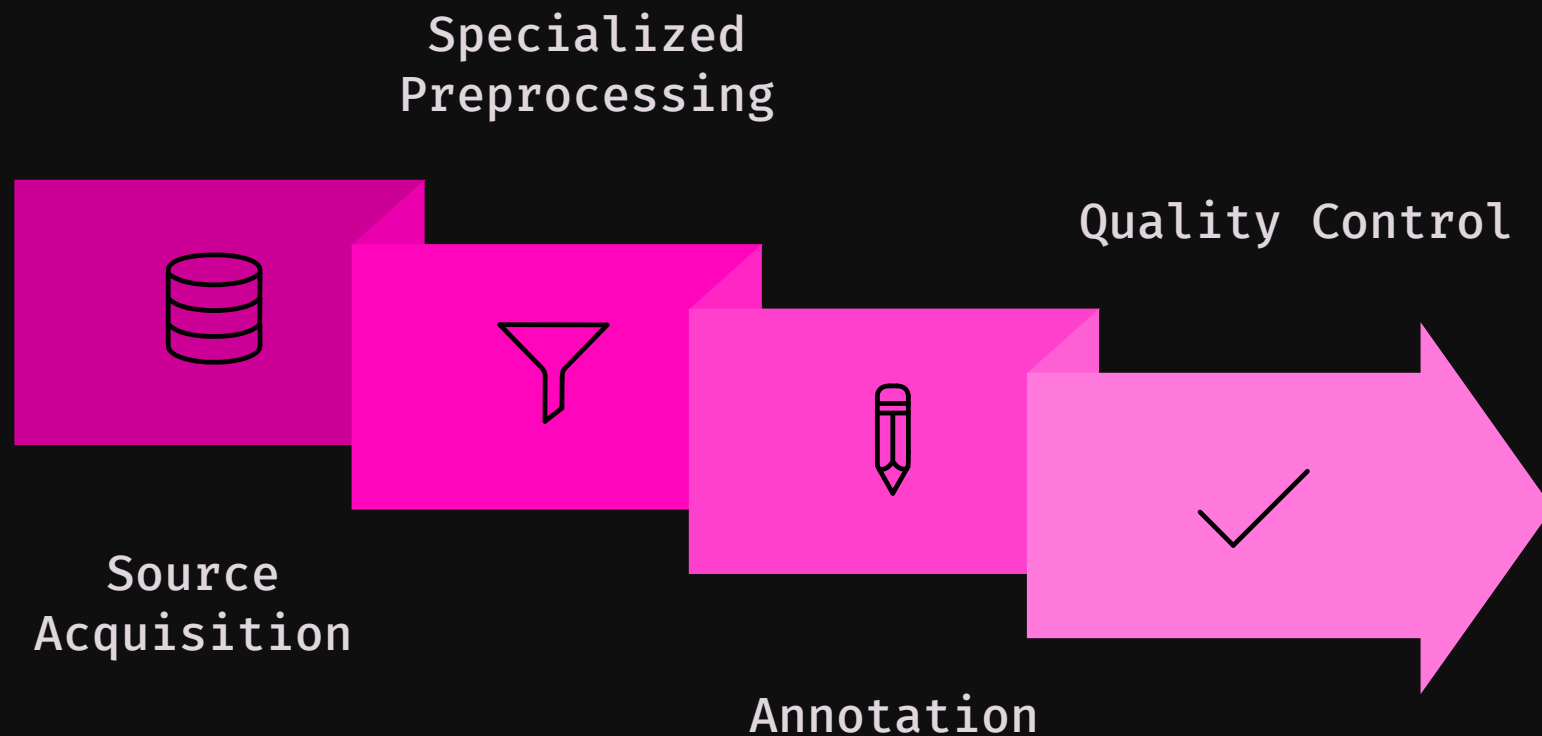
Proposed Framework Flowchart

Purpose: To visually map the rigorous, step-by-step process required to construct the high-quality Urdu text summarization dataset, from data collection to validation.



Framework Component 2: Annotation Process

The annotation phase is critical for generating reliable gold-standard summaries. Our framework emphasizes **strict quality control** through a systematic and well-defined process.



During annotation, human experts are tasked with creating ideal summaries. Each summary is then evaluated using a 1-5 Likert scale, assessing aspects like fluency, coherence, and accuracy.

Likert Scale Scoring: Annotators rate summary quality from 1 (Poor) to 5 (Excellent), providing granular feedback crucial for iterative refinement and consistency.

ROGUE Scores: (Recall-Oriented Understudy for Gisting Evaluation) - measures overlap between the system's summary and the generated summary. ROGUE-1 is unigram overlap, ROGUE-2 is bigram overlap. ROGUE-L is longest common subsequence (a good proxy for the summary's overall structure and preservation of the main sentence meaning). high ROGUE score = dataset utility validated.

A line drawing of two men in white lab coats shaking hands. The man on the left has light hair and is wearing a white lab coat over a dark shirt and tie. The man on the right has dark hair and is also wearing a white lab coat over a dark shirt and tie. Between them is a small square icon containing a line graph with an upward arrow and a circle at the end of the line.

$$\kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}$$

$\bar{P}$ = The average observed agreement among annotators.

$\bar{P}_e$ = The agreement expected by chance.

Inter-Annotator Agreement (IAA) Visualization

The diagram shows three annotators, each with a sample sentence and a score of 5/5 (Relevant). The scores are used for Fleiss' Kappa (k) Calculation, resulting in a Dataset Reliability Check with $k \approx 0.7$ (Goal Achieved).

Annotator 1
5 کنا جن با کلیر بکھ گھوڑ کنا مچ کپڑکھ
Sample Sentence: The quick brown fox jumps...
Score: 5/5 (Relevant)

Annotator 2
5 کنا جن با کلیر بکھ گھوڑ کنا مچ کپڑکھ
Sample Sentence: The quick brown fox jumps...
Score: 5/5 (Relevant)

Annotator 3
5 کنا جن با کلیر بکھ گھوڑ کنا مچ کپڑکھ
Sample Sentence: The quick brown fox jumps...
Score: 5/5 (Relevant)

Fleiss' Kappa (k) Calculation

Dataset Reliability Check
 $k \approx 0.7$
(Goal Achieved)

Framework Component 3: Specialized Preprocessing

Given Urdu's unique linguistic characteristics, standard preprocessing techniques are insufficient. Our framework incorporates **specialized steps** to prepare the text for effective summarization.

Normalization for Nastaliq

- Addressing ligatures and contextual character forms unique to Nastaliq script.
- Standardizing character representations to reduce variability.
- Ensuring consistent encoding and rendering for computational processing.

Morphological Analysis

- Breaking down words into their constituent morphemes (roots, stems, affixes).
- Enabling better feature engineering by understanding word structure.
- Crucial for handling Urdu's rich inflectional system.

These steps are vital to accurately capture the semantic and syntactic information necessary for robust summarization models, especially in a low-resource context.

Framework Component 4: Scale and Size

Building a foundational dataset requires a strategic approach to scale, balancing initial feasibility with long-term utility. Our framework defines clear size targets to ensure progressive development.

 500+

Pilot Dataset

Minimum pairs for initial validation and framework refinement.

 5,000–10,000

Target Dataset

Optimal pairs for robust model training and evaluation.

Strategic Scaling: The pilot phase (500+ pairs) is designed for rigorous testing of the framework and annotation guidelines. Upon successful validation, the target scale (5,000–10,000 pairs) provides sufficient data volume for developing and comparing sophisticated summarization algorithms.